System Validation Report

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1. Purpose and scope

This report records whether the Hostel Management System meets the requirements in the SRS. It covers the system test (ST) cases in the updated test case document, the non-functional and security requirements, and the requirement traceability matrix (RTM) built from every test case result.

2. Summary

Run date	11 November 2026
Code under test	feature/part3-testing (based on main at c7f9932 plus feature/student-home)
Environment	Windows 11, Python 3.11.9, Django 5.2.19, SQLite 3, coverage.py 7.17.2
Requirements validated	30 of 31 (19 functional, 6 non-functional, 6 security)
System test cases	16 of 17 passed; 1 not run (ST_UX_01)
Open defects	0 Critical, 0 High, 0 Medium; 2 Low (section 6)
Exit criteria (STP 5)	Met, except ST_UX_01 which is scheduled for the UAT demo

3. How system tests were executed

The STP planned system tests as manual browser runs, later automated with Selenium. They were automated with the Django test client instead: each ST test drives the real URLs, views, forms, templates and database end to end, exactly as a browser would, and runs on every commit in CI. Two cases cannot run inside the test suite and were executed separately: ST_REL_01 (restart, run as two processes on a file database) and ST_UX_01 (needs real users).

4. System test results

ID	Description	Requirement	Expected	Actual	Result
ST_AUTH_01	Login lands on the right page for each role	HMS-F-001, F-002	Admin sees dashboard; student sees room availability; wrong password shows an error	As expected; 3 automated test(s) passed	**Pass**
ST_STU_01	Admin registers, edits and searches a student	HMS-F-003, F-004, F-017	Success messages; updated phone visible; search finds the student	As expected; 3 automated test(s) passed	**Pass**
ST_ROOM_01	Admin sets up a hostel with rooms	HMS-F-006, F-007, F-009	All three rooms listed with correct capacity and availability	As expected; 1 automated test(s) passed	**Pass**
ST_AVL_01	Student view updates after admin allocates	HMS-F-008	Available beds reduced by 1 after the allocation	As expected; 1 automated test(s) passed	**Pass**
ST_ALLOC_01	Admin fills a room until it is full	HMS-F-010, F-011	First two succeed; third shows capacity error; room shows 'Full'	As expected; 1 automated test(s) passed	**Pass**
ST_ALLOC_02	Admin modifies, cancels and checks history	HMS-F-013, F-014, F-018	Message at each step; history shows all three actions	As expected; 1 automated test(s) passed	**Pass**
ST_CICO_01	Full stay: allocate, check in, check out	HMS-F-015, F-016	Buttons enabled only when allowed; final status COMPLETED; bed free	As expected; 1 automated test(s) passed	**Pass**
ST_DASH_01	Dashboard matches the database	HMS-F-019	Every value on the page equals a direct database count	As expected; 1 automated test(s) passed	**Pass**
ST_PERF_01	90% of pages within 3 s	HMS-NF-001	90th percentile <= 3 s; 0 server errors	160 requests on 500 students / 100 rooms: p90 31 ms, max 46 ms, 0 errors	**Pass**

ID	Description	Requirement	Expected	Actual	Result
ST_REL_01	Data survives restart	HMS-NF-002	All records present and unchanged after restart	Process 1 wrote student PES1UG24AM777, room REL-R-1, ACTIVE checked-in allocation and 2 history rows, then exited. Process 2 re-opened the database: all records, check-in and password hash unchanged.	**Pass**
ST_SEC_01	Passwords stored hashed	HMS-SR-003	Values start with pbkdf2_sha256$; no plaintext password	As expected; 2 automated test(s) passed	**Pass**
ST_SEC_02	Broken access control check	HMS-SR-002, SR-006	Every admin URL and foreign record returns 403 or 404	As expected; 1 automated test(s) passed	**Pass**
ST_SEC_04	Injection, bad input and CSRF	HMS-SR-004	Rejected or stored as text; no 500; no CSRF token gives 403	As expected; 3 automated test(s) passed	**Pass**
ST_UX_01	Usability walk-through	HMS-NF-004	All tasks finished without help; messages rated clear	Needs three students who did not build HMS. To be run during the UAT demo.	**Not run**
ST_MAINT_01	Modular structure	HMS-NF-005	One app per module; full suite passes after a change	As expected; 1 automated test(s) passed	**Pass**
ST_DATA_01	Capacity and availability stay consistent	HMS-NF-006	ACTIVE <= capacity; availability = capacity - ACTIVE	As expected; 1 automated test(s) passed	**Pass**
ST_CI_01	CI pipeline runs the suite with coverage gate	HMS-NF-005	Workflow passes on Python 3.11 and 3.13 with coverage >= 80%	Workflow commands (makemigrations --check, coverage run --branch, coverage report --fail-under=80) pass locally: 202 tests, 99%. Confirm the green run on the GitHub Actions tab.	**Pass (local)**

5. Non-functional and security validation

Requirement	Evidence	Verdict
HMS-NF-001 Performance	ST_PERF_01: 90th percentile 31 ms against a 3 s target; room list query count constant as rooms grow (TC_PERF_01).	Met
HMS-NF-002 Reliability	ST_REL_01: records survive a restart; failed actions roll back (atomic services).	Met
HMS-NF-003 / SR-001 Mandatory login	IT_AUTH_02: every protected URL redirects to login.	Met
HMS-NF-004 Usability	TC_UX_01 automated checks pass; ST_UX_01 user walk-through not yet run.	Partly validated
HMS-NF-005 Maintainability	ST_MAINT_01: one app per module; CI runs the whole suite.	Met
HMS-NF-006 Data integrity	ST_DATA_01: 200 random actions (seed 42), capacity rule held after every action; UT_ALLOC_05 concurrency.	Met
HMS-SR-002 / SR-006 Access control	IT_AUTH_03, IT_STU_03, ST_SEC_02, IT_SEC_02.	Met
HMS-SR-003 Password hashing	ST_SEC_01: pbkdf2_sha256 hashes, unique salts.	Met
HMS-SR-004 Input validation	ST_SEC_04, IT_SEC_01: SQL injection, script tags, 500-character strings, missing CSRF token (403).	Met
HMS-SR-005 Sessions	IT_AUTH_04: logout flushes the session.	Met
Deployment check (STP 5.1)	manage.py check --deploy with DEBUG=False: 2 warnings, W004 (HSTS not set) and W008 (SSL redirect off). Both need HTTPS, which this project does not deploy.	Noted (D-05)

6. Open defects and known issues

ID	Severity	Description	Status
D-04	Low	No logs/hms.log is configured (SAD 3.4), so the log search step of ST_SEC_01 does not apply. Services still call the logger, with IDs only.	Open
D-05	Low	Deploy check warnings W004 and W008 (HTTPS settings).	Open, HTTPS out of scope

7. Requirement traceability matrix

Requirement	Name	Test cases	Status
HMS-F-001	User authentication	UT_AUTH_01, UT_AUTH_02, ST_AUTH_01	Validated
HMS-F-002	Role-based access control	UT_AUTH_04, IT_AUTH_03, ST_AUTH_01, IT_CO_02, IT_DASH_01	Validated
HMS-F-003	Student registration	UT_STU_01, UT_STU_02, ST_STU_01	Validated
HMS-F-004	Update student details	UT_STU_04, ST_STU_01	Validated
HMS-F-005	View student profile	IT_STU_02	Validated
HMS-F-006	Hostel management	UT_HOS_01, ST_ROOM_01	Validated
HMS-F-007	Room inventory and management	UT_ROOM_01, UT_ROOM_02, UT_ROOM_03, ST_ROOM_01	Validated
HMS-F-008	Room availability tracking	UT_AVL_02, UT_AVL_03, IT_AVL_01, ST_AVL_01	Validated
HMS-F-009	Room search and view	IT_ROOM_02, ST_ROOM_01	Validated
HMS-F-010	Room allocation	UT_ALLOC_01, UT_ALLOC_02, IT_ALLOC_01, ST_ALLOC_01	Validated
HMS-F-011	Prevent over-allocation	UT_ALLOC_03, UT_ALLOC_04, UT_ALLOC_05, IT_ALLOC_02, ST_ALLOC_01	Validated
HMS-F-012	View allocation status	IT_STAT_01	Validated
HMS-F-013	Modify room allocation	UT_MOD_01, UT_MOD_02, ST_ALLOC_02	Validated
HMS-F-014	Cancel room allocation	UT_CAN_01, ST_ALLOC_02	Validated
HMS-F-015	Student check-in	UT_CI_01, UT_CI_02, UT_CI_03, IT_CI_01, ST_CICO_01	Validated
HMS-F-016	Student check-out	UT_CO_01, UT_CO_02, UT_CO_03, IT_CO_01, IT_CO_02, IT_CO_03, ST_CICO_01	Validated
HMS-F-017	Search student records	UT_STU_05, ST_STU_01	Validated
HMS-F-018	Allocation history log	UT_HIS_01, IT_HIS_01, ST_ALLOC_02	Validated
HMS-F-019	Administrative summary dashboard	UT_DASH_01, IT_DASH_01, ST_DASH_01	Validated
HMS-NF-001	Performance (90% of pages within 3 s)	ST_PERF_01	Validated
HMS-NF-002	Reliability and persistence	ST_REL_01	Validated
HMS-NF-003	Security (mandatory authentication)	IT_AUTH_02	Validated
HMS-NF-004	Usability and clear messages	ST_UX_01	Partly validated
HMS-NF-005	Maintainability (modular apps)	ST_MAINT_01, ST_CI_01	Validated
HMS-NF-006	Data integrity of capacity	UT_AVL_03, UT_ALLOC_05, ST_DATA_01	Validated
HMS-SR-001	Mandatory authentication	IT_AUTH_02	Validated
HMS-SR-002	Role-based authorisation	UT_AUTH_04, IT_AUTH_03, ST_SEC_02, IT_SEC_02	Validated
HMS-SR-003	Password hashing	ST_SEC_01	Validated
HMS-SR-004	Input validation and sanitisation	UT_STU_03, ST_SEC_04, IT_SEC_01	Validated
HMS-SR-005	Session management and logout	IT_AUTH_04	Validated
HMS-SR-006	Student data privacy	IT_STU_03, ST_SEC_02	Validated

HMS-NF-004 is partly validated only because ST_UX_01 is still to be run; its automated usability checks pass.